



HITESH PATIL

+919325467007 | hiteshvinodpatil@gmail.com | www.linkedin.com/in/hitesh-patil-1326842a8

PROFESSIONAL SUMMARY

Software development enthusiast with a strong focus on problem-solving and intelligent systems. Skilled in Full-Stack Web Development, Database Management, Data Structures & Algorithms (DSA), and System Design. Completed coursework in AI/ML and Deep Learning, with a passion for building scalable, real-world software solutions

TECHNICAL SKILLS

Languages: Python, Java (DSA), JavaScript, SQL

Web: HTML, CSS, Bootstrap, Tailwind CSS, React.js, Angular, Node.js, Express.js

Databases: MySQL, MongoDB

AI / ML: Machine Learning, Deep Learning, TensorFlow, Keras, NumPy, Pandas, Scikit-learn

Gen AI Technology: LangChain, LangGraph, RAG, ChromaDB, OpenAI API, Gemini API, FastAPI, Multi-Agent Systems, Vector Databases

Tools & Platforms: Git, GitHub, VS Code, Postman, Power BI, Excel

Other: Data Structures & Algorithms, Problem Solving

EXPERIENCE

EV_Tracker Project — EV Battery Lifecycle Management & Governance System (IIT Mandi Project)

- Developed a full-stack EV Battery Lifecycle Management System to track batteries from manufacturing, usage, and ownership to final recycling.
- Implemented secure JWT Authentication and Role-Based Access Control (RBAC) for Manufacturer, Owner, Recycler, and Admin roles.
- Built RESTful APIs using Node.js and Express.js and integrated MySQL to manage battery details, lifecycle stages, and user information.
- Designed Audit Logging to record battery registration, status updates, ownership changes, and recycling activities, ensuring complete traceability and transparency.

PROJECTS

CodeSage AI — Multi-Agent Code Review & Intelligent Software Quality Platform

- Developed CodeSage AI, an AI-powered code review platform using Python, FastAPI, React.js, LangGraph, and ChromaDB to automate repository analysis.
- Implemented a multi-agent architecture with specialized AI agents to detect bugs, security vulnerabilities, performance issues, code quality problems, documentation gaps, and unit test coverage.
- Built a RAG (Retrieval-Augmented Generation) pipeline using Sentence Transformers and ChromaDB to enable repository-aware AI chat with semantic code search.
- Integrated GitHub repository analysis, automated code indexing, and an interactive dashboard that generates quality scores, findings, and actionable recommendations for developers.

LifeLineZero — AI-Powered Pre-Disaster Management & Early Warning System

- Developed an AI-powered disaster management platform with JWT Authentication, emergency incident reporting, AI-based disaster severity prediction, risk assessment, and evacuation planning to support civilians and rescue authorities.
- Built scalable REST APIs using Node.js and Express.js, integrated MongoDB, and implemented core modules including SOS Beacon tracking, Rescue Command Dashboard, Resource Tracker, Multilingual Emergency Alerts, and Offline Survival Mode.
- Integrated AI-powered What-If Scenario Simulation, Risk & Readiness Engine, and geolocation-based analysis to predict disaster impact, identify high-risk areas, and recommend safe evacuation routes for emergency response.
- Implemented advanced features including 2D-to-3D Building Visualization using Three.js for rescue planning and a Blockchain-based Donation System to ensure transparent and secure disaster relief management.

EDUCATION

B.Tech – Mechanical Engineering | Vishwakarma Inst. of Info. Technology, Pune 2023 – 2027 | CGPA: 8.45

Higher Secondary(12th–Science) | ASC College, Chopda, Jalgaon 2022 – 2023 | 80.00%

SSC (10th) | VVC High School, Chopda, Jalgaon 2020 – 2021 | 86.20%

CERTIFICATIONS

- Web Development — YouTube
- Generative AI — YouTube
- Java Programming & Data Structures and Algorithms — Coursera
- AI/ML, Deep Learning & TensorFlow — Coursera

ACHIEVEMENTS & HACKATHONS

- Disaster Management Hackathon — Built a real-time emergency response platform; ranked Top 10 out of 250+ teams.
- Smart India Hackathon (SIH) — Developed a Market Access App for Farmers; selected as SIH Finalist.
- Finalist at DYDPDU Hackathon for SatyaCheck, an AI-based Fake News Detection system.
- Solved 230+ DSA problems on LeetCode (leetcode.com/u/iv5q8Rkq7f/).